Natural Language Processing

First Semester, 1402-1403

Hands-on Exam

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Teaching Assistants

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16th of Day, 1402

The duration of the exam is 180 minutes

Exam Rules

- Your entire screen during the exam must be recorded and delivered on a flash drive at the end of the exam.
- It is permissible to use the codes of exercises and workshops that you entered yourself or that were provided to you during the workshop.
- Only and only the following online resources can be used. Using any other source is considered cheating.

o https://huggingface.co/docs

o https://docs.python.org

o https://numpy.org/doc

o https://pytorch.org/docs/ ۲.۰

o https://nlp.stanford.edu/projects/glove

- Copying any piece of code from anywhere is considered cheating.
- Any contact with anyone using any method, including git is considered cheating and will lead to failure in the course.
- The exam is through GradeScope and it closes at 12pm. You must send the notebook run without clearing the output with the same name. For the parts related to the assignments submit only modified files. In both notebooks and exercises, tag your codes , with the hashtag #NLPEXAM . Finally, upload all the files at once to GradeScope..

Question ۱ (۱۲ points)

In theQ1_embedding notebook functions , Complete parts a, b, c and d and take the corresponding tests.

Question ۲ (۱۳ points)

In parser_model.py file froma assignment 3 in the model designed for ,dependency parsing put a hidden layer along with the ReLU activation function after the first layer and before the last layer, and set the number of neurons in this layer equal to hidden_size. Make the necessary changes for this. (Note that you may change more than one function)

Question ۳ (۱۵ points)

In assignment 5 by changing the code in part e, use the additive attention mechanism instead of the attention mechanism used, run part d and report the results. The formula of the additive attention mechanism is as follows:

$$f_{att}(\mathbf{h}_i, \mathbf{s}_j) = v_a^T \tanh(\mathbf{W}_a[\mathbf{h}_i; \mathbf{s}_j])$$

where v and W are learnable parameters.

Question ۴ (۱۶ points)

In notebook Q4_huggingface_torch Answer the questions 1 to 8 and perform the tests for each one.

Question ۵ (۱۴ points)

In Q5_transformer notebook , instead of using sinusoidal positional encoding use learnable , position encoding and just run the first part of the code again with the changes applied (no need to run thehomework part).

Question ۶ (۱۰ points)

In theQ6_social_media notebook, determine the sentiment of the first 200 data points of the Reuters dataset and create a plot for it.

Question ۷ (۲۰ points)

In Q7_VQA notebook, set the number of classes to 5. Instead of using Adam as an optimizer, use AdamW . Use resnet50 instead of resenet18 (may need to resize the hidden_dim variable). Then, according to the link below, freeze the parameters of the image_feature_extractormodel:

https://jimmy-shen.medium.com/pytorch-freeze-part-of-the-layers-4554105e03a6

Next,write a scheduler (CyclicLR - exp_range) and set the following parameter s for it:

https://www.kaggle.com/code/isbhargav/guide-to-pytorch-learning-rate-scheduling

base_lr=0.001, max_lr=0.1, step_size_up=71, mode="exp_range", cycle_momentum=False

Just pay attention to update the learning rate after doing the evaluation in each epoch because otherwise, the evaluation will not be fair.

Question ۸ (۱۰ points) (Extra)

In the Q8_interpretable notebook first run the first cell. Then change the context so that the answer is alice .